Machine Learning Model Development & Evaluation Plan

WEEK 3 ASSIGNMENT

ACADEMIC PERFORMANCE FORECASTING PIPELINE

1. Problem Definition

- **Project Title:** Student Academic Performance Prediction
- **Objective:** Build a predictive machine learning pipeline to estimate final academic performance based on study habits, attendance rates, continuous assessment scores, and engagement metrics.
- **Relevance:** Enables educational institutions to identify at-risk students early in the term, facilitating timely and targeted academic interventions.

2. Data Preprocessing & Feature Engineering

- **Data Cleaning:** Handling missing numerical data using median imputation to mitigate outlier impact, and mode imputation for categorical attributes.
- **Feature Scaling:** Applying StandardScaler to normalize numerical features for scale-sensitive algorithms (e.g., Logistic Regression, SVM).
- **Feature Engineering:** Constructing derived attributes (such as Attendance-to-Study-Hour ratio) and encoding categorical variables using One-Hot Encoding.

Key Challenge & Solution:

Challenge: Class imbalance in student risk categories.
Solution: Applying SMOTE (Synthetic Minority Over-sampling Technique) to rebalance the training set synthetic instances.

3. Machine Learning Workflow

END-TO-END PIPELINE ARCHITECTURE

4. Model Selection & Training Strategy

- **Candidate Models Evaluated:**
  - **Linear/Logistic Regression:** Establishes a transparent baseline for simple linear relationships.
  - **Decision Trees:** Captures non-linear dependencies with clear operational interpretability.
  - **Random Forest / Gradient Boosting:** Ensemble architectures that maximize predictive accuracy and reduce variance/overfitting.

- **Selected Primary Model:** Random Forest Classifier / Regressor due to high performance, handles non-linear relationships well, and demonstrates robustness against noisy data.
- **Hyperparameter Tuning:** Utilizing GridSearchCV and RandomizedSearchCV to optimize max depth, estimator count, and split criteria.

5. Performance Evaluation & Validation Strategy

- **Evaluation Metrics:**
  - **Regression Approach:** Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.
  - **Classification Approach:** Accuracy, Precision, Recall, F1-Score, and ROC-AUC curve evaluation.
- **Validation Strategy:** Implementing 5-Fold Cross-Validation to ensure robust model generalization and eliminate data leakage.

6. Model Deployment & Maintenance Plan

- **Serialization:** Exporting full processing and model pipelines using joblib or pickle.
- **API Integration:** Serving the serialized model via Flask or FastAPI endpoints for integration with student dashboards.
- **Monitoring & Maintenance:** Conducting periodic performance audits and scheduled retraining with fresh semester data to prevent data drift.